

Within-system hydrogen-permeation relaxation and identifiability

The original workbook contains two hydrogen-permeation current traces at 293 K under cathodic-current increases from 0.5 to 1 and from 5 to 10 mA/cm². It does not identify independent specimens. The task is therefore forward extrapolation within the measured system, using the first 80% of each trace for development and the last 20% for reserved confirmation.

Eleven substantive scientific attempts were completed, plus one invalid technical implementation retained separately. The selected activated-breakthrough surrogate attains 0.674 μA mean trace MAE on the reserved tails, compared with 4.278 μA for the polynomial control. The result supports a compact late-time forecast under these conditions. It does not identify a new physical law or a unique material diffusivity.

The equation is $I(t)=I_0+(A_0+A_1z) \exp[-(\tau \exp(kz)/t)^\beta]$, where z identifies the known high-current condition, t and τ are seconds, I_0 is the initial three-point calibration, A_0 and A_1 are μA, and k and β are dimensionless. Condition-dependent effective timing improves prediction, but boundary entry, trapping and diffusion remain confounded.

For the slab competitors, the dimensionless response is

$$S(x) = 1 + 2 \sum_{n=1}^{40} (-1)^n \exp[-n^2 \max(x, 0.001)].$$

The function erfc is the complementary error function. A dimensional lag error invalidated the first slab implementation; its files and correction are preserved, and it is excluded from model selection.

Experimental scope and evaluation

Forward extrapolation within two linked permeation traces. Parameters are learned only from earlier chronological blocks; confirmation is the last 20% of each trace. Known imposed current step and initial three-point current calibration are permitted.

Calibration: Mean first 3 native current observations per trace; all candidate families receive identical values. Fit parameters may use development prefixes only.

Validation unit: Two chronological traces with no independent specimen identifiers; nonoverlapping temporal blocks are scoring units, not independent specimens. All trace linkage is explicit.

Report both trace-tail errors; no population interval or independent-specimen significance test.

Target: measured anodic permeation current (μA). Errors use μA. The selected reference is **activated_breakthrough**. Selection was frozen before confirmation; diagnostic winners are not substituted afterward.

Predictor	Unit
t	s since third baseline observation
offset	μA; mean first 3 observations
drive	mA/cm ² ; imposed cathodic-current step
high	binary known high-current condition

Matched comparison

Model	Development MAE	Reserved MAE	Reserved worst-group MAE
fixed_diffusion	20.852	26.1811	47.9259
flexible	11.9525	4.27767	6.18262
persistence	27.6682	31.2127	48.0014
shared_boundary_response	7.31915	8.2512	8.56553
fick_slab_corrected	10.3538	5.92083	9.29837
surface_saturation	5.06106	2.36247	4.36316
trapping_two_timescales	7.04349	8.34861	12.2288
occupancy_dependent_transpo rt	4.78902	2.31295	3.63909
shared_fick_separate_amplitud es	3.34284	0.854128	1.39143
distributed_trapping	2.52537	0.887883	1.52371
finite_capacity_trapping	5.11432	3.85729	5.91291
activated_breakthrough	1.51247	0.674135	0.876482
semi_infinite_arrival	3.42019	3.42719	5.12817
common_breakthrough_scale	3.77347	1.26306	1.79216

The primary error is mean group MAE. RMSE and signed bias are complementary; rows within a group do not establish independent replication. No accuracy percentage or industrial tolerance is invented.

Selected equation and coefficients

$$\hat{J}(t) = J_0 + (a_0 + a_1 H) \exp \left[- \left(\frac{\tau e^{kH}}{t} \right)^\beta \right]$$

J0 is the allowed initial current offset in microamperes; H is the declared high-current-condition indicator. Time t and tau are in seconds; a 0 and a 1 are in microamperes, while k and beta are dimensionless. The effective timescale is not a uniquely identified diffusivity.

Coefficient	Frozen value
a 0	15.6672957
a 1	36.5888402
tau	2087.48867
k	-0.57031044
beta	1.59867711

All comparator expressions, numerical guards and full-precision values remain in EQUATIONS.md, rules.json and run.py.

Applicability and limitations

- The two traces do not establish independent-specimen transfer. Temperature is 293 K; nominal sheet thickness 1.2 mm.

- Native time zero is used as reported; true boundary concentration and its step timing are not independently measured.
- An effective relaxation scale must not be relabeled a unique lattice diffusivity because trapping, boundary kinetics and baseline uncertainty can be confounded.

First two observations per sheet were inspected for schema and belong to permitted initial calibration. No reserved tail current was inspected before freeze.

Reproduction

Run `python run.py` in this final package to verify hashes and reproduce every saved prediction. Use `python run.py --inputs new.csv --output predictions.csv --model reference` only with the declared inputs and units. Numerical prediction evaluation does not certify novelty or mechanism. Source anchors and reserved observations are separate from predictor inputs.

Source: <https://data.mendeley.com/datasets/tnsgf7v76z/1>. See source/units audit and prior-art records in research history.